Category

Mortgage-Backed Security (MBS) Instrument and Market Features

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RESIDENTIAL MORTGAGE-BACKED SECURITIES (RMBS)

Learning Outcome

- describe types and characteristics of residential mortgage-backed securities, including mortgage pass-through securities and collateralized mortgage obligations, and explain the cash flows and risks for each type

This section starts with a discussion of different RMBS types, to continue with an overview of mortgage pass-through securities and non-agency RMBS. It concludes with collateralized mortgage obligations.

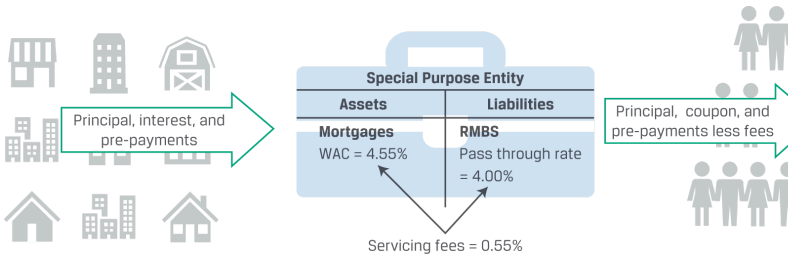
A **mortgage pass-through security** is a security created when mortgage lenders pool mortgages together and sell securities to investors. The cash flow from the mortgage pool—monthly payments of principal, interest, and prepayments—are "passed through" to the security holders. A pool can consist of several thousand or only a few mortgages. When a mortgage is used as collateral for a mortgage pass-through security, the mortgage is said to be securitized.

Mortgage Pass-Through Securities

The cash flows of a mortgage pass-through security depend on the monthly cash flows of the underlying pool of mortgages. The amount and the timing of the cash flows from the pool of mortgages cover both the cash flow passed through to the security holders and administrative charges for servicing the pool; the structure is depicted in Exhibit 2.

Exhibit 2:

Mortgage Pass-Through



Note: WAC = weighted average coupon rate.

These administrative charges include collecting monthly payments from borrowers, forwarding proceeds to owners of the loan, sending payment notices to borrowers, maintaining records of the outstanding mortgage balance, initiating foreclosure proceedings if necessary, and providing tax information to borrowers when applicable. Additionally, the issuer or financial guarantor of the mortgage pass-through security may charge fees for guaranteeing the issue. All these fees are typically calculated as a portion of the mortgage rate.

A comparison of MBS and RMBS can be found in Exhibit 3.

Exhibit 3:

Comparison of MBS and RMBS

collateral	commercial mortgages	securities
Subtype	▪ RMBS ▪ CMBS	▪ Mortgage pass-through securities ▪ Collateralized mortgage obligations (CMOs), including sequential pay, planned amortization class (PAC), and support tranches ▪ Agency RMBS ▪ Non-Agency RMBS

A mortgage pass-through security's coupon rate is called the **pass-through rate** and is lower than the weighted average mortgage rate earned on the underlying pool of mortgages because of administrative charges. The pass-through rate that the investor receives is said to be "net interest" or "net coupon."

The mortgages in a securitization pool are heterogeneous: outstanding principal, interest rates, and maturities vary across the mortgages. Consequently, for each mortgage pass-through security, a **weighted average coupon rate (WAC)** and a **weighted average maturity (WAM)** are determined. The WAC is calculated by weighting the mortgage rate of each mortgage in the pool by the percentage of the outstanding mortgage balance relative to the outstanding amount of all the mortgages in the pool. Similarly, the WAM is calculated by weighting the remaining number of months to maturity of each mortgage in the pool by the outstanding mortgage balance relative to the outstanding amount of all the mortgages in the pool.

Exhibit 4 shows the five mortgages underlying a specific mortgage-backed security.

Exhibit 4:

Weighted Average Coupon Rate and Weighted Average Maturity

Mortgage	Interest rate (*i*)	Beginning Balance (BB)	Current Balance (CB)	Original Term (months)	Number of Months to Maturity (MM)
A	2.50%	EUR300,000	EUR238,000	240	180
B	3.30%	EUR420,000	EUR380,000	600	480
C	2.80%	EUR100,000	EUR87,000	288	240
D	4.00%	EUR280,000	EUR132,000	360	120
E	3.70%	EUR350,000	EUR312,000	384	312
		EUR1,450,000	EUR1,149,000		

The weighted average coupon rate for this MBS is calculated as follows:

$$WAC = i_A\left(\frac{CB_A}{\Sigma_{CB}}\right)+i_B\left(\frac{CB_B}{\Sigma_{CB}}\right)+i_C\left(\frac{CB_C}{\Sigma_{CB}}\right)+i_D\left(\frac{CB_D}{\Sigma_{CB}}\right)+i_E\left(\frac{CB_E}{\Sigma_{CB}}\right)$$

$$WAC = 2.50\%\left(\frac{238,000}{1,149,000}\right)+3.30\%\left(\frac{380,000}{1,149,000}\right)+2.80\%\left(\frac{87,000}{1,149,000}\right)$$
$$+4.00\%\left(\frac{132,000}{1,149,000}\right)+3.70\%\left(\frac{312,000}{1,149,000}\right)$$

WAC = 3.29%.

The weighted average maturity for this MBS is calculated as follows:

$$WAM = 180\left(\frac{238,000}{1,149,000}\right)+480\left(\frac{380,000}{1,149,000}\right)+240\left(\frac{87,000}{1,149,000}\right)+120\left(\frac{132,000}{1,149,000}\right)+312\left(\frac{312,000}{1,149,000}\right)\text{month}$$

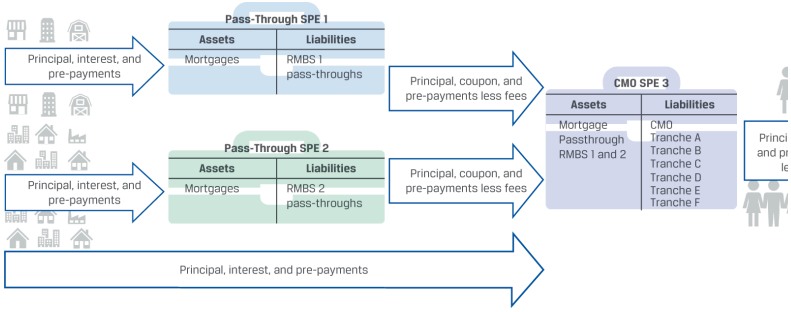
WAM =313 months.

Collateralized Mortgage Obligations (CMOs)

Collateralized mortgage obligations securitize mortgage pass-through securities or multiple pools of loans. CMOs are structured to redistribute the cash flows to different bond classes or tranches and create securities that have different exposures to prepayment risk. Exhibit 5 depicts a simplified CMO structure where the cash flows are pooled and redistributed across the different tranches.

Exhibit 5:

Collateralized Mortgage Obligation (CMO)



▶ Expand / collapse Extended Description

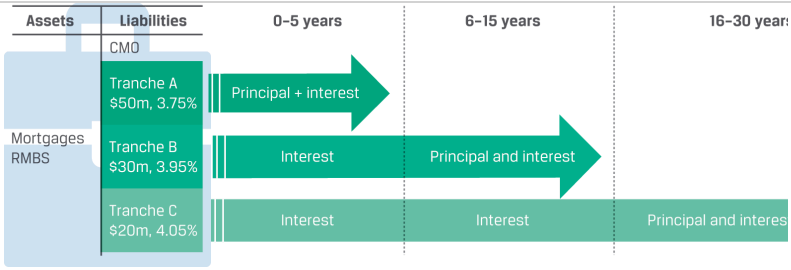
The tranching structure in the CMOs cannot eliminate prepayment risk, but it can redistribute the prepayment risk across the different tranches, insulating some tranches more than others. The CMO tranche structures reduce the uncertainties of the size and timing of payments investors receive. The features of these tranches are designed to specifically meet the various needs of institutional investors. Across the range of CMO structures, the more senior a tranche is, the less exposure it has to prepayment risk and default risk.

Sequential-Pay CMO

In sequential-pay CMO structures, each tranche would be retired sequentially. This is an example of time tranching. One class begins to receive principal payments from the underlying securities only after the principal on any previous class has been completely paid off and retired. All principal payments are made to Tranche A until the principal balance for Tranche A is zero. After Tranche A is paid off, the principal payments go to Tranche B until the principal balance for Tranche B is zero. This continues until all the tranches are repaid.

To illustrate a sequential-pay CMO, let us use a hypothetical transaction with the terms outlined above. The collateral for this CMO is the mortgage pass-through security described in Exhibit 2, where the total par value of the collateral is USD100 million, the pass-through coupon rate is 4%, the WAC is 4.55%, and the WAM is 360 months (about 30 years). From this collateral, three tranches are created, as shown in Exhibit 6. The coupon rate varies with the differences in maturity and reflects the term structure of interest rates, among other factors. Exhibit 6 shows the resulting structure.

Exhibit 6:



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Some protection against prepayment risk is provided for each tranche. The protection arises because prioritizing the distribution of principal (that is, establishing the payment rule for the principal repayment) effectively protects the shorter-term tranche (A in this structure) against extension risk. This protection comes from the longer-term tranches B and C. At the same time, Tranches B and C benefit because they are provided protection against contraction risk from Tranche A. Thus, the sequential-pay CMO structure allows investors concerned about extension risk to invest in Tranche A, and those concerned about contraction risk to invest in Tranches B and C.

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Although the payment rules for the distribution of the principal repayments are known, the actual principal repayment amount in each month is not, as it depends on the actual prepayment rate of the collateral. In the sequential-pay CMO structure, the tranches have average lives that are shorter or longer than the collateral, thereby attracting investors who have preferences for different average lives. A pension fund that expects a significant increase in the number of retirements after several years may opt for a tranche with a longer average life, such as Tranche B. An endowment that looks to finance large building projects over the next few years may choose a tranche with lower average life, such as Tranche A. And long-term investors willing to assume more risk may select Tranche C.

Knowledge Check

1. Referring to the sequential-pay CMO in Exhibit 6, match each investor type to the tranche that *most closely* fits its needs.

Investor Types:

a. A pension fund looking for higher returns in exchange for accepting more risk
b. A mutual fund seeking safe returns for a small portion of its portfolio.
c. An insurance company that sells annuities to clients expected to retire starting in approximately fifteen years

Tranche	Investor Type
Tranche A Notes Principal Amount: USD50 million Coupon Rate: 3.75% Maturity Date: [Five Years from Settlement Date]	
Tranche B Notes Principal Amount: USD30 million Coupon Rate: 3.95% Maturity Date: [Fifteen Years from Settlement Date]	
Tranche C Notes Principal Amount: USD20 million Coupon Rate: 4.05% Maturity Date: [Thirty Years from Settlement Date]	

Hide Solution

Solution:

USD50 million Coupon Rate: 3.75% Maturity Date: [Five Years from Settlement Date]	
Tranche B Notes Principal Amount: USD30 million Coupon Rate: 3.95% Maturity Date: [Fifteen Years from Settlement Date]	c) An insurance company that sells annuities to clients expected to retire starting in approximately fifteen years.
Tranche C Notes Principal Amount: USD20 million Coupon Rate: 4.05% Maturity Date: [Thirty Years from Settlement Date]	a) A pension fund looking for higher returns in exchange for accepting more risk.

Other CMO Structures

The cash flows from an underlying mortgage pool can be structured in additional ways. The structures below describe various tranches that are created through time tranching (Z-tranche and residual tranche) or by splitting up cash flows from the pool.

- *Z-Tranches* do not pay interest payments until a pre-set date, when both principal and accrued interest payments start. During the accrual period at each payment date, the principal value of the Z-tranche is credited by the stated coupon rate. Typically, Z-tranches are the last tranche in a series of sequential or PAC and companion tranches. A Z-tranche benefits the other tranches because it frees up cash flows that other tranches can distribute. At the same time, holders of the Z-tranche do not face reinvestment risk when market yields decline. Z-tranches usually have average lives in excess of 20 years. This makes these tranches risky and hard to value. Z-tranches are also known as accretion bonds or accrual bonds.
- *Principal-Only (PO) securities* pay only the principal repayments from the pool. These securities can be created either from mortgage pass-throughs or as a tranche in a CMO. Because PO investors receive the face value of their securities through scheduled principal payments and prepayments, the value of these securities is very sensitive to prepayment rates and interest rates. With falling interest rates or when prepayments accelerate, the value of the PO will increase.
- *Interest-Only (IO) securities* pay their holders only the interest payments from the pool. IO securities are a companion of the PO securities in a CMO transaction. These securities have no face or par value. With increased prepayments, the cash flows paid to the IO investors decline. That is why investors use IOs to hedge their portfolios against interest rate risk.
- *Floating-Rate tranches* carry interest rates that are linked to an index or a market reference rate. The interest rates are variable and are often subject to both a cap and a floor rate. These tranches are impacted by how interest rate movements affect prepayment rates. Floating-rate tranches can also be structured as inverse floaters, where the interest paid changes in the opposite direction of the change in interest rates. They are used to hedge interest rate risk in portfolios.
- *Residual tranches* collect any remaining cash flow from the pool after all the obligations to the other tranches are met. These investments are appropriate for investors that can assume the risk and hedge it. Such investors include hedge funds and long-term institutional investors. Banks typically avoid such investments due to capital requirements.

make scheduled and fixed principal payments over a predetermined time period to their investors if the prepayment levels in the pool are within a certain maximum and minimum range. If the prepayment rate is within the specified range, all prepayment risk is absorbed by the support tranche.

Question Set

The following table is used for questions 1, 2, and 3:

Shown below are five mortgages that comprise the entire pool in a mortgage-backed security.

Mortgage	Interest Rate (*i*)	Beginning Balance (BB)	Current Balance (CB)	Original Term (Months)	Number of Months to Maturity
A	2.00%	CAD200,000	CAD83,000	420	140
B	3.00%	CAD300,000	CAD193,000	360	192
C	4.00%	CAD400,000	CAD315,000	360	240
D	5.00%	CAD500,000	CAD357,000	240	144
E	6.00%	CAD600,000	CAD548,000	420	324
		CAD2,000,000	CAD1,496,000		

1. The WAC for this security is *closest* to:

A. 4.00%.
B. 4.50%.
C. 4.73%.

Hide Solution

Solution:

The correct answer is C. The weighted average coupon is calculated by weighting the mortgage rate of each mortgage in the pool by the percentage of the outstanding mortgage balance relative to the outstanding amount of all the mortgages in the pool, as shown in the equation below.

$$WAC = i_A\left(\frac{CB_A}{\Sigma_{CB}}\right)+i_B\left(\frac{CB_B}{\Sigma_{CB}}\right)+i_C\left(\frac{CB_C}{\Sigma_{CB}}\right)+i_D\left(\frac{CB_D}{\Sigma_{CB}}\right)+i_E\left(\frac{CB_E}{\Sigma_{CB}}\right)$$

$$WAC = 2.00\%\left(\frac{83,000}{1,496,000}\right)+3.00\%\left(\frac{193,000}{1,496,000}\right)+4.00\%\left(\frac{315,000}{1,496,000}\right)$$
$$+5.00\%\left(\frac{357,000}{1,496,000}\right)+6.00\%\left(\frac{548,000}{1,496,000}\right)$$

WAC = 4.73%.

A is incorrect because it is calculated as a straight average of the five interest rates of the five mortgages in the pool.

B is incorrect because it is calculated by weighting the mortgage rate of each mortgage in the pool by its percentage of the pool's total beginning mortgage balance, not by its percentage of the pool's current mortgage balance.

2. Referring to the Summary of Terms for the Sequential-Pay CMO Structure with Three Tranches Government Housing Trust, pool 917-37 shown in Exhibit 3, identify the investor *most likely* to select each tranche.

Investor	Tranche
1. A fund that is expected to provide retirement benefits beginning in about ten years	A. Tranche A

3. A foundation that expects to award sizeable grants for medical research within the next few years and wants to avoid extension risk	C. Tranche C

Hide Solution

Solution:

1. B is correct. A fund that expects to provide retirement benefits in about ten years may opt for a tranche, such as Tranche B. (In the sequential-pay CMO structure, the tranches have average lives that are shorter or longer than the underlying collateral, thereby attracting investors who have preferences for different average lives.)
2. C is correct. A long-term fund seeking high returns would *most likely* be willing to assume more risk in order to achieve those higher returns. Tranche C has the longest term and is the riskiest of the three tranches.
3. A is correct. A foundation that requires funding within the next few years and thus wants to avoid extension risk (the risk that borrowers will slow down payments and not meet scheduled payment timelines) *most likely* would opt for Tranche A.

3. *Determine the correct answers to complete the following sentences:*

The tranching structure in the CMOs cannot eliminate _____ but can redistribute it across the different tranches, insulating some more than others. Across the range of CMO structures, the more senior a tranche is, the less exposure it has to _____ and _____.

Hide Solution

Solution:

The tranching structure in the CMOs cannot eliminate *prepayment risk* but can redistribute it across the different tranches, insulating some more than others. Across the range of CMO structures, the more senior a tranche is, the less exposure it has to *prepayment risk* and *default risk*.

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MO Q2, there's no pool 917-37 in Exhibit 3, I think the question is referring to an exhibit that's not in the text?

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